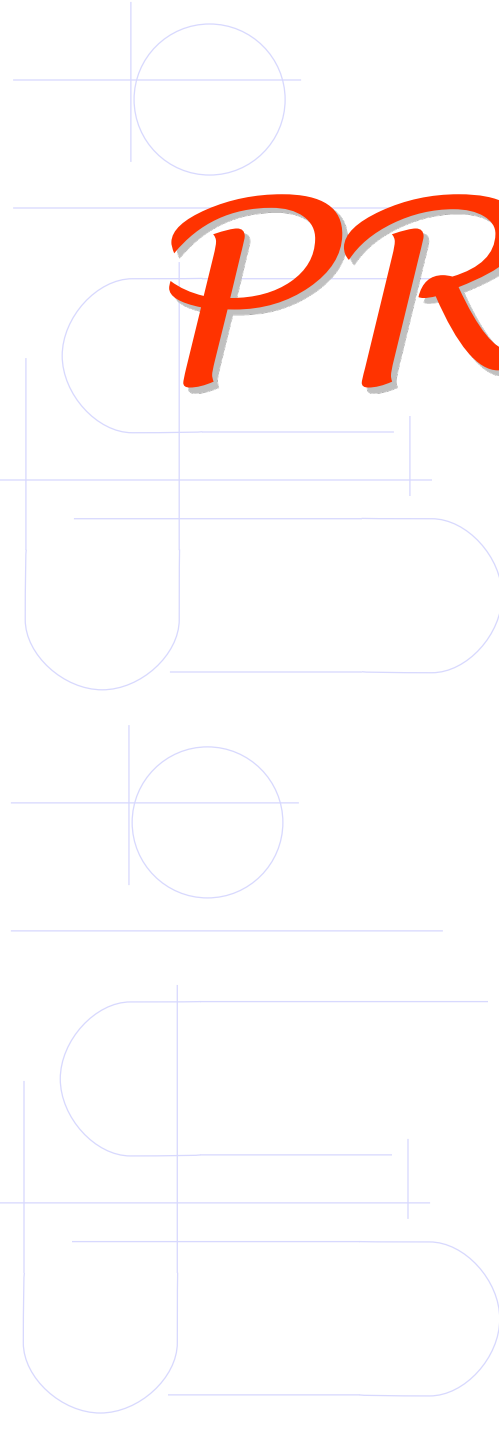




PRACTICAS

Fase I

Fase I

ARCHIVOS

$f_1.jpg$

$f_2.jpg$

$f_3.mp3$

$f_4.mp3$

$f_5.mp4$

$f_6.mp4$

CLAVES
128 bits

k_{f_1}

k_{f_2}

k_{f_3}

k_{f_4}

k_{f_5}

k_{f_6}

CIFRADO
AES128

$$E_{k_{f_1}}^{AES}(f_1) = c_1$$

$$E_{k_{f_2}}^{AES}(f_2) = c_2$$

$$E_{k_{f_3}}^{AES}(f_3) = c_3$$

$$E_{k_{f_4}}^{AES}(f_4) = c_4$$

$$E_{k_{f_5}}^{AES}(f_5) = c_5$$

$$E_{k_{f_6}}^{AES}(f_6) = c_6$$

EN CLARO



Fase I

ARCHIVOS

$f_1.jpg$
 $f_2.jpg$
 $f_3.mp3$
 $f_4.mp3$
 $f_5.mp4$
 $f_6.mp4$

CLAVES 128 bits

$E_{K_u}^{RSA}(k_{f_1})$
 $E_{K_u}^{RSA}(k_{f_2})$
 $E_{K_u}^{RSA}(k_{f_3})$
 $E_{K_u}^{RSA}(k_{f_4})$
 $E_{K_u}^{RSA}(k_{f_5})$
 $E_{K_u}^{RSA}(k_{f_6})$

CIFRADO AES128

$E_{k_{f_1}}^{AES}(f_1) = c_1$
 $E_{k_{f_2}}^{AES}(f_2) = c_2$
 $E_{k_{f_3}}^{AES}(f_3) = c_3$
 $E_{k_{f_4}}^{AES}(f_4) = c_4$
 $E_{k_{f_5}}^{AES}(f_5) = c_5$
 $E_{k_{f_6}}^{AES}(f_6) = c_6$

Usuario: u

RSA

K_u clave pública de u

k_u clave privada de u

PROTEGIDAS

EN CLARO



Fase I

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$f_5.mp4$

$f_6.mp4$

CLAVES

128 bits

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$$E_{K_u}^{RSA}(k_{f_3})$$

$$E_{K_u}^{RSA}(k_{f_4})$$

$$E_{K_u}^{RSA}(k_{f_5})$$

$$E_{K_u}^{RSA}(k_{f_6})$$

CIFRADO

AES128

$$E_{k_{f_1}}^{AES}(f_1) = c_1$$

$$E_{k_{f_2}}^{AES}(f_2) = c_2$$

$$E_{k_{f_3}}^{AES}(f_3) = c_3$$

$$E_{k_{f_4}}^{AES}(f_4) = c_4$$

$$E_{k_{f_5}}^{AES}(f_5) = c_5$$

$$E_{k_{f_6}}^{AES}(f_6) = c_6$$

Usuario: u

RSA

K_u clave pública de u

k_u clave privada de u

EN CLARO

Usuario: u

Password:

p_u password de u

¿128 bits?

PROTEGIDA

$$E_{p_u}^{AES}(k_u)$$

Se puede hacer uso de una función hash



Fase I

ARCHIVOS

$f_1.jpg$

$f_2.jpg$

$f_3.mp3$

$f_4.mp3$

$f_5.mp4$

$f_6.mp4$

CLAVES

128 bits

$$E_{K_u}^{RSA}(k_{f_1})$$

$$E_{K_u}^{RSA}(k_{f_2})$$

$$E_{K_u}^{RSA}(k_{f_3})$$

$$E_{K_u}^{RSA}(k_{f_4})$$

$$E_{K_u}^{RSA}(k_{f_5})$$

$$E_{K_u}^{RSA}(k_{f_6})$$

CIFRADO

AES128

$$E_{k_{f_1}}^{AES}(f_1) = c_1$$

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$$E_{k_{f_5}}^{AES}(f_5) = c_5$$

$$E_{k_{f_6}}^{AES}(f_6) = c_6$$

Usuario: u

RSA

K_u clave pública de u

k_u clave privada de u

$$E_{p_u}^{AES}(k_u)$$

Usuario: u

Password:

p_u password de u

Se puede hacer uso de una función hash

$$\text{SHA-3}^{256}(p_u)$$

